

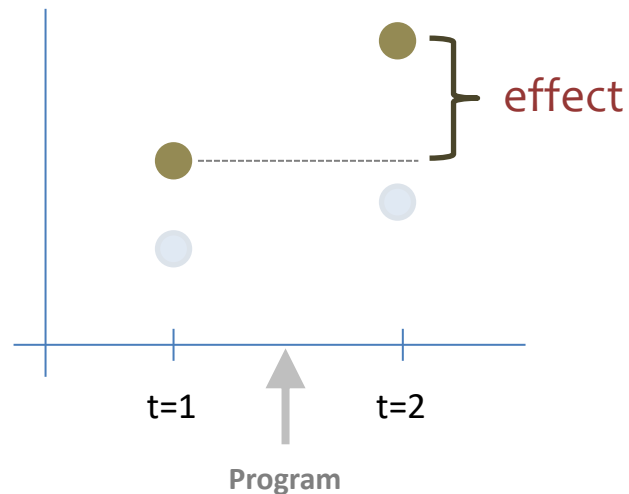
REGRESSION DISCONTINUITY DESIGN

3 VALID COUNTERFACTUALS:

Each variety of counterfactual has a different formula for the program effect estimate.

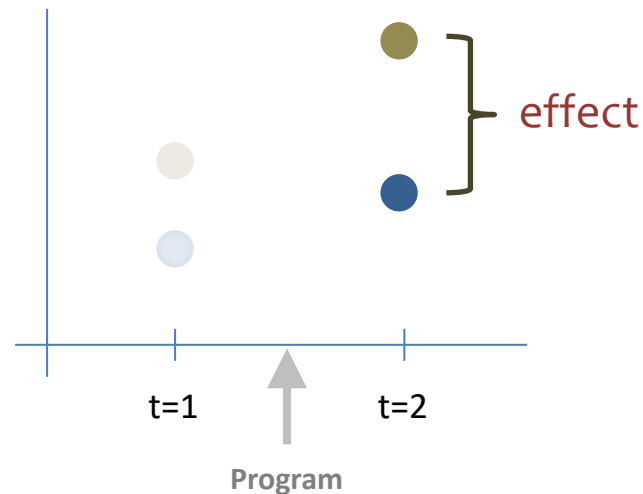
PRE-POST REFLEXIVE
Estimator

$$\text{effect} = T2 - T1$$



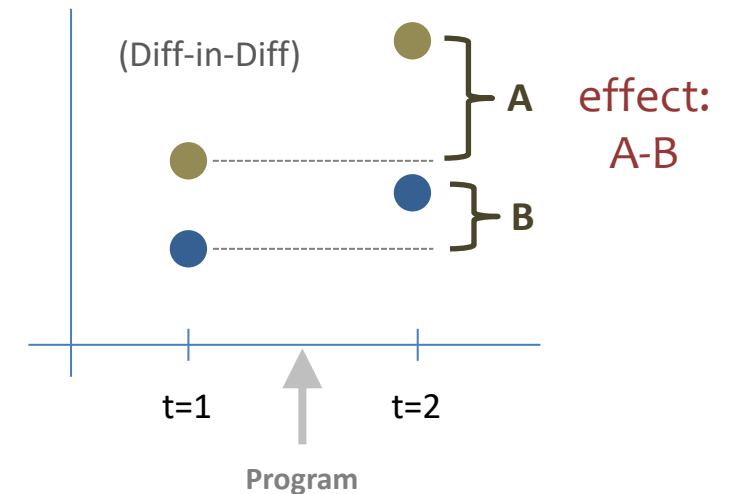
POST-ONLY
Estimator

$$\text{effect} = T2 - C2$$



PRE-POST W COMPARISON
Estimator

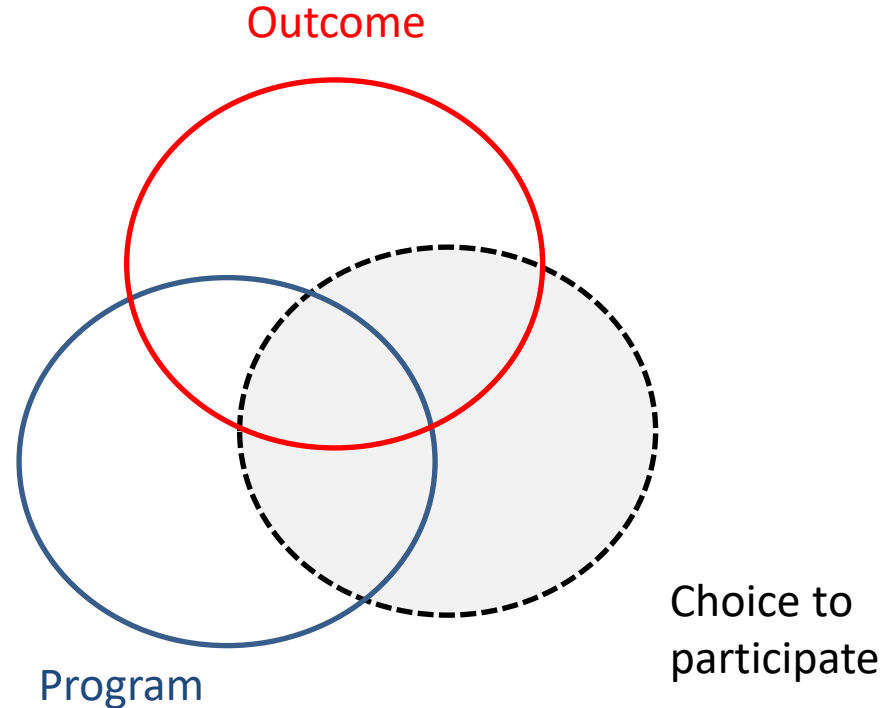
$$\text{effect} = (T2 - T1) - (C2 - C1)$$



● Treatment Groups, **T1**=before, **T2**=post-program measure

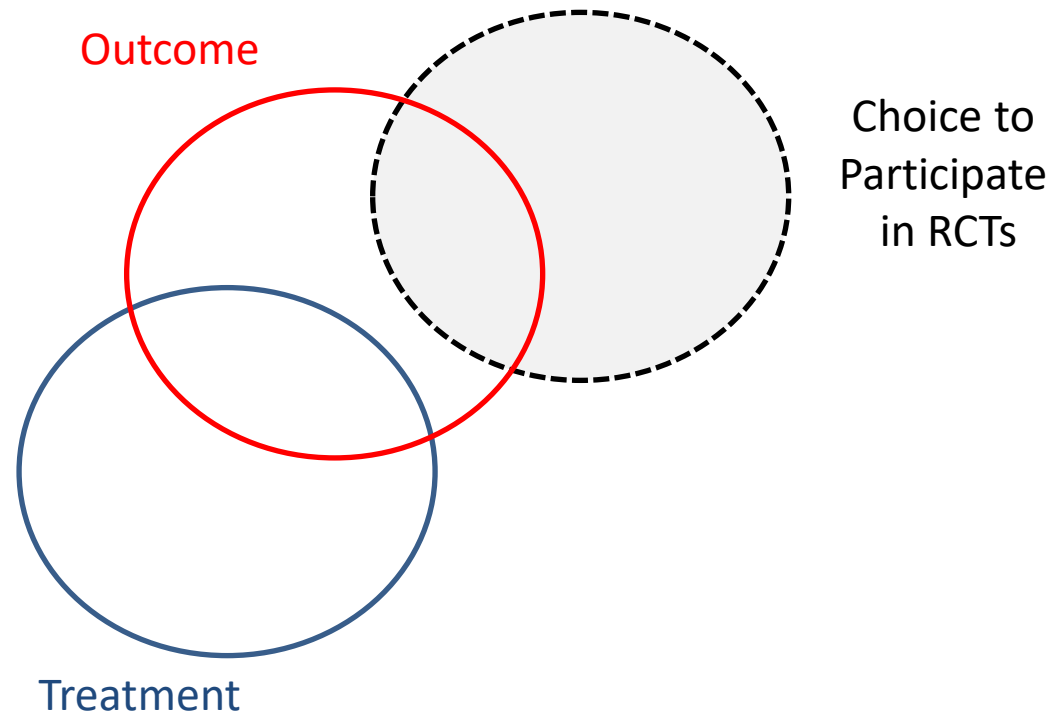
● Control Groups, **C1**=before, **C2**=post-program measure

The “selection” problem

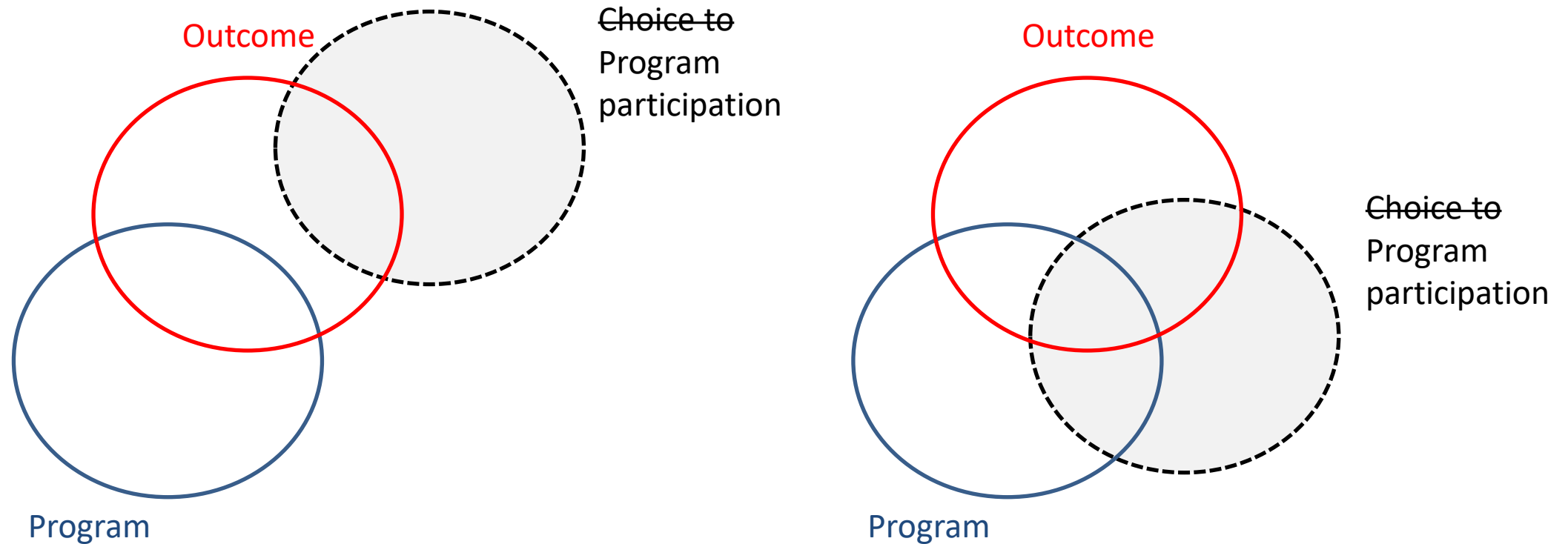


When happens when people are given the choice to participate in a program? When do problems arise?
How do we estimate program impact in this case?

Solution: randomization

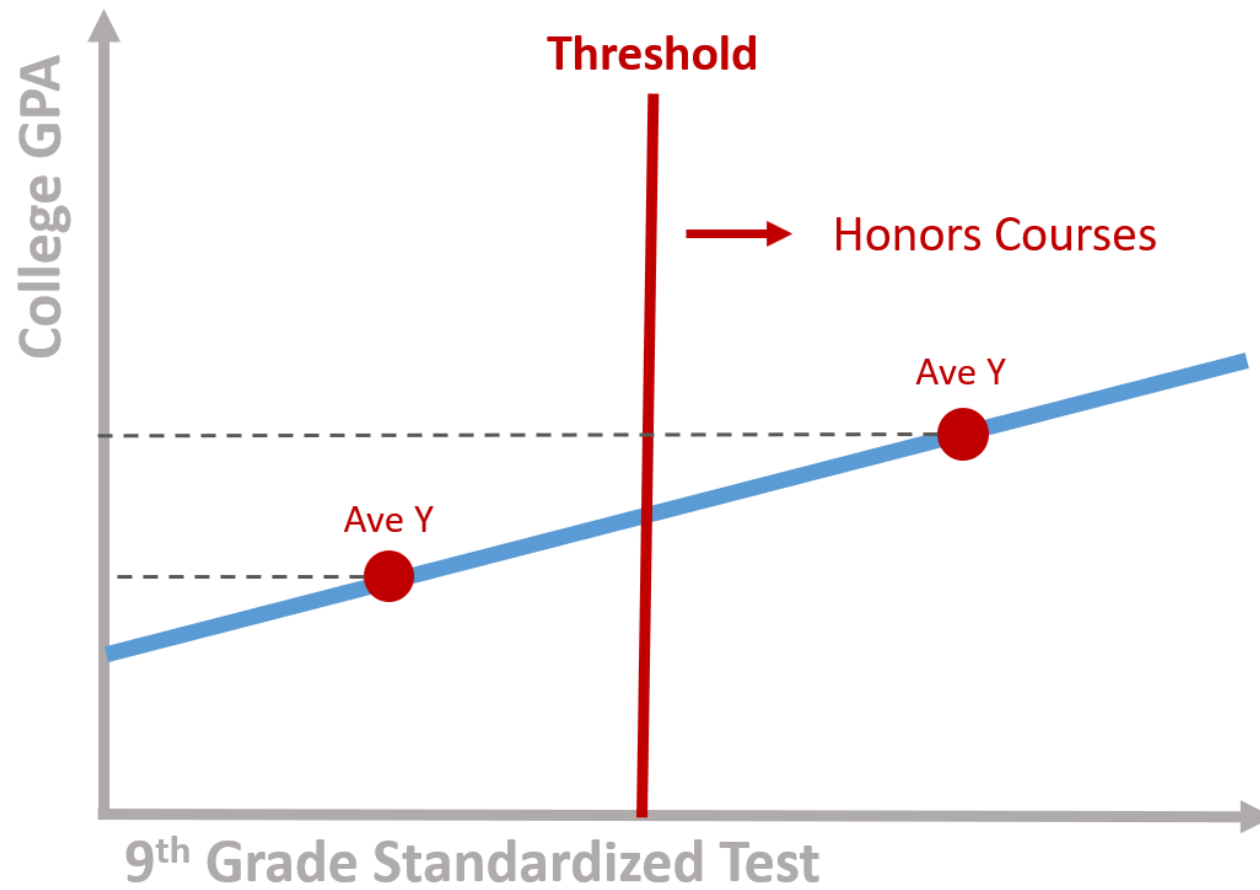


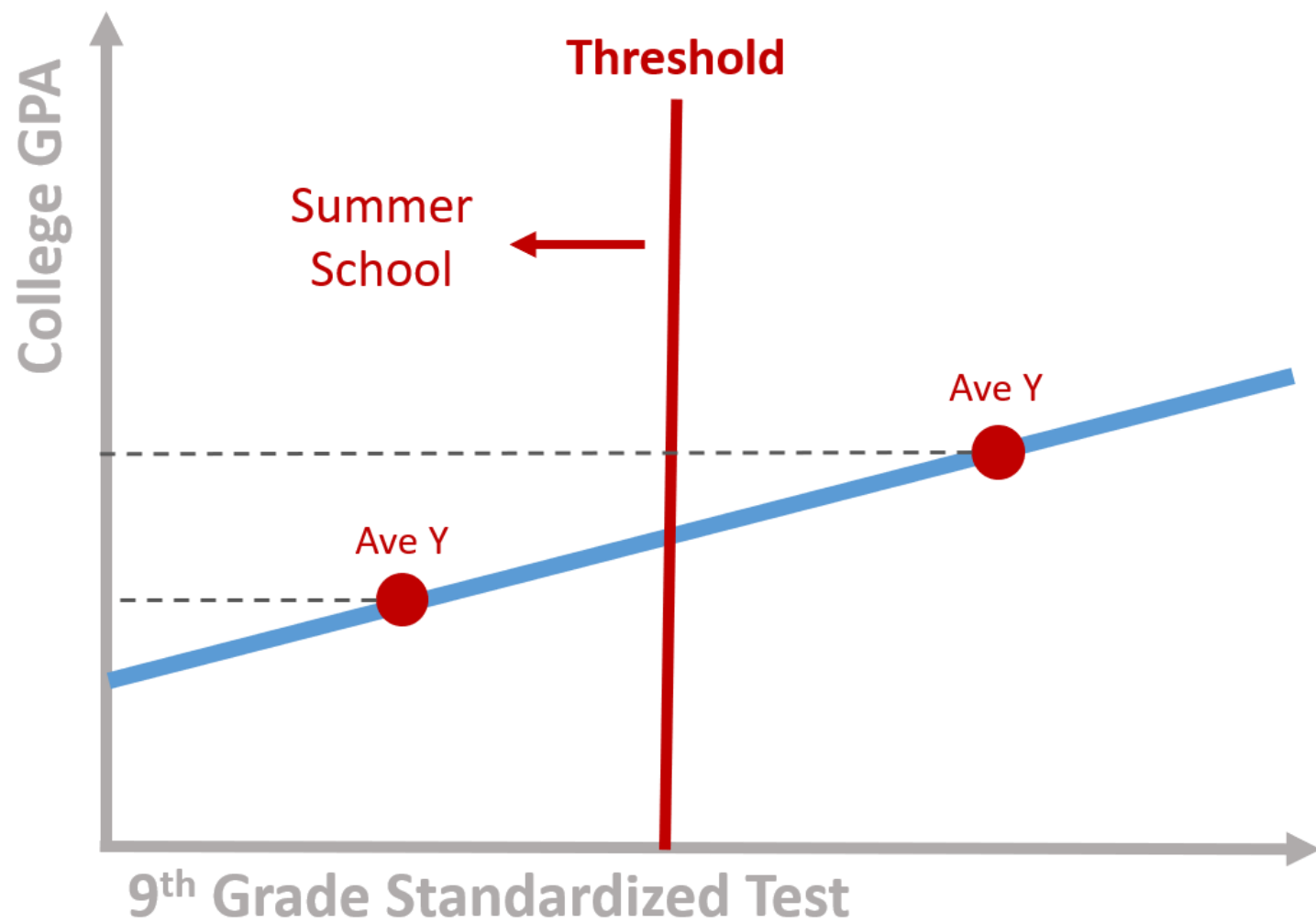
The “selection” problem when participation is mandatory but you need to qualify

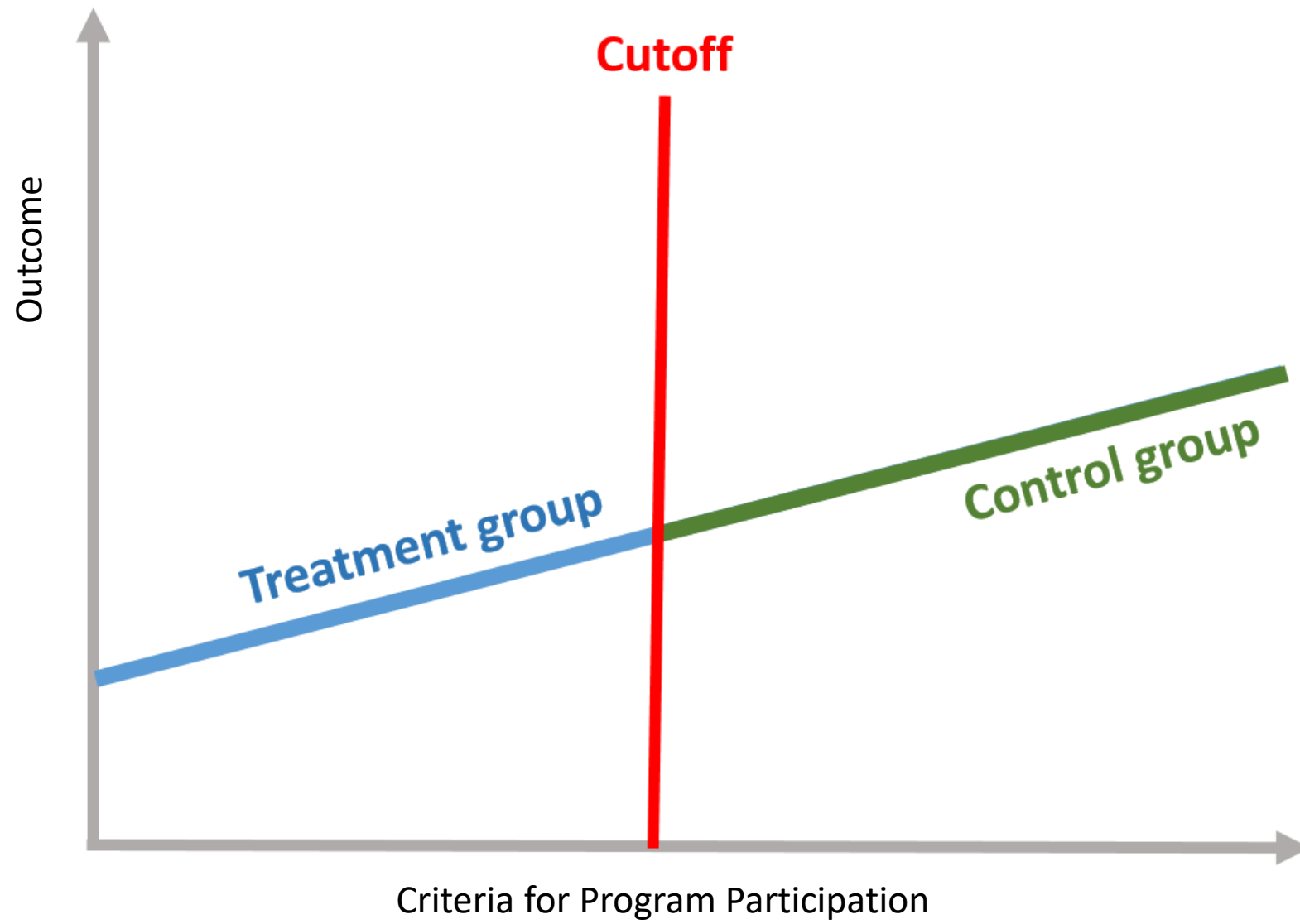


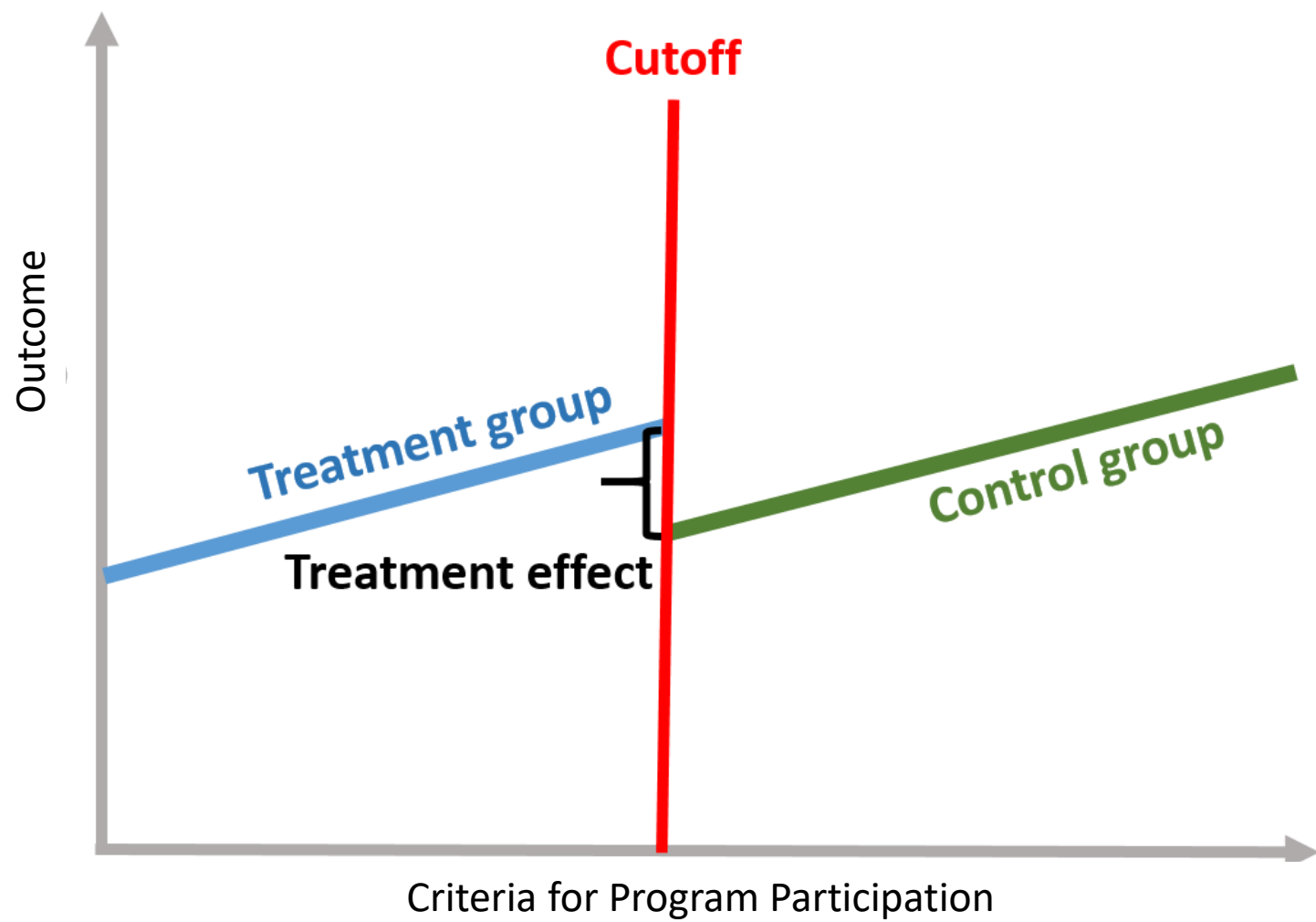
What about means-tested programs?
E.g. Medicare, housing vouchers, food stamps,
honors classes, college scholarships, summer school

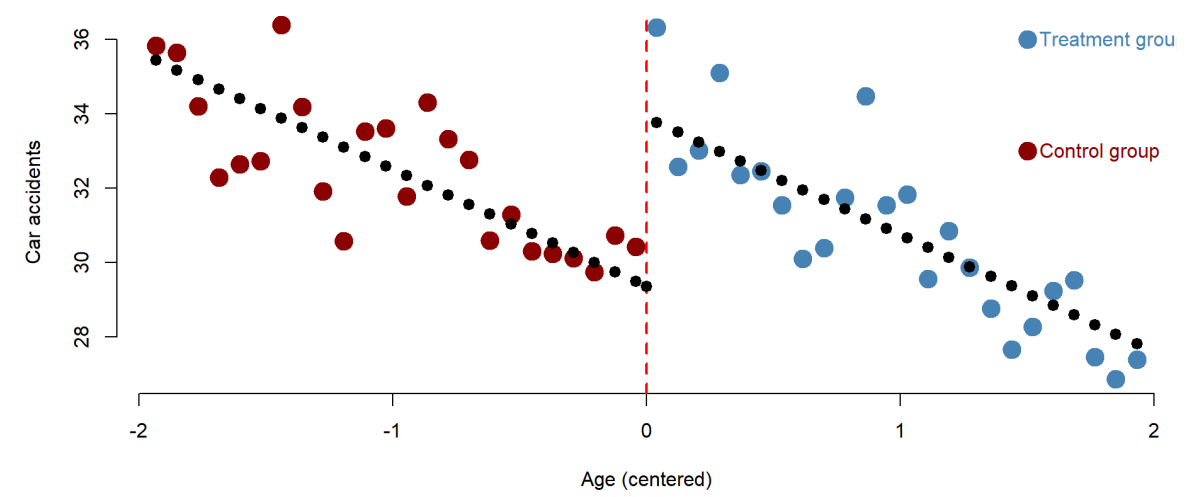
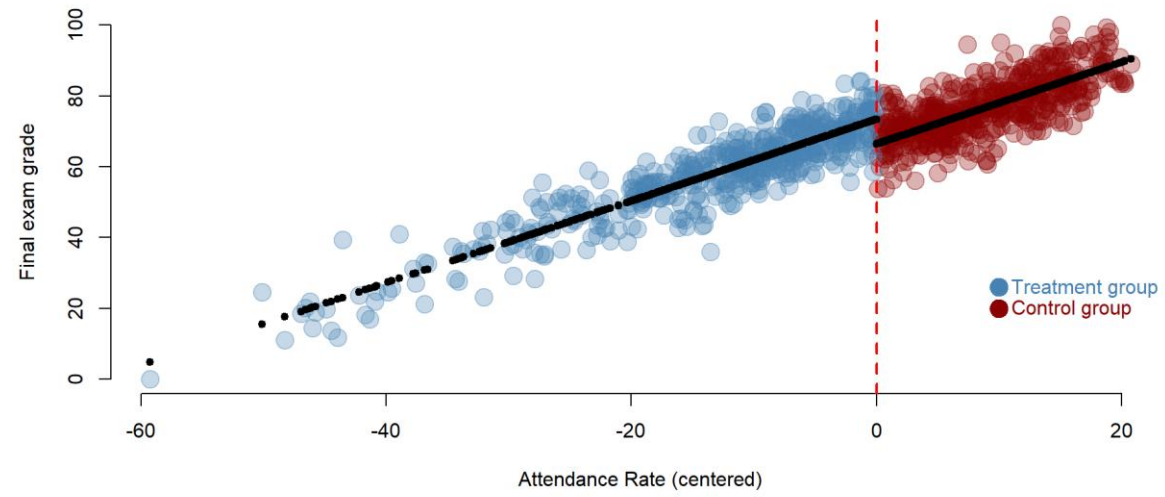
REGRESSION DISCONTINUITY DESIGN











$$Y = \beta_0 + \beta_1 * Treatment + \beta_2 * Ratingvariable_c + e \quad (1.1)$$

Where:

Y is our outcome variable;

Treatment is a dummy variable indicating the treatment (=1) and the control (=0) group;

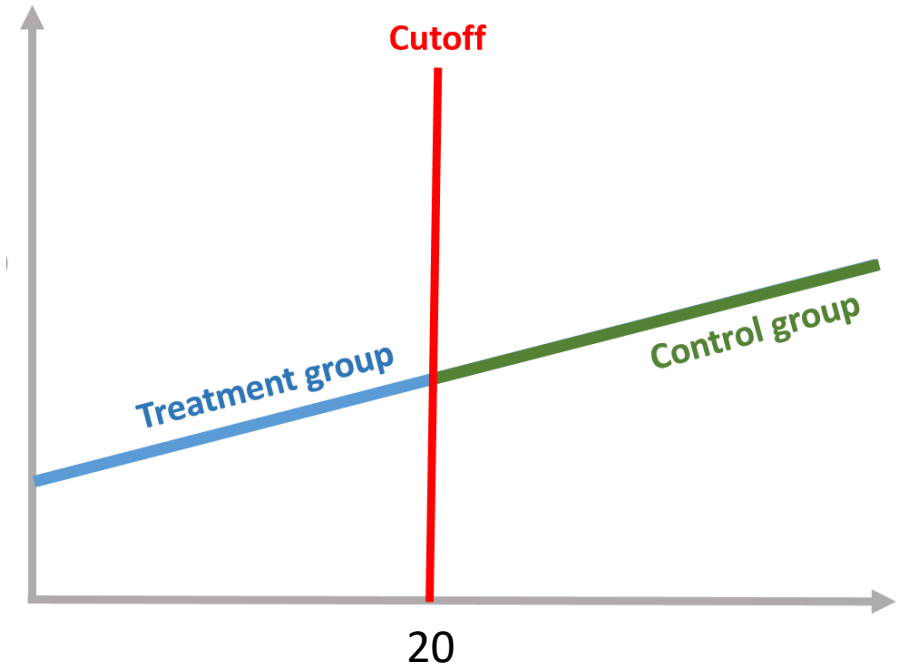
Rating variable_c is the rating variable centered at the cutoff point. To center the variable, we just need to subtract the cutoff score from each individual score. The cut-off point will become the new intercept (since cut-off - cut-off = 0), which it will help to interpret the results.

Centered
On Cutoff

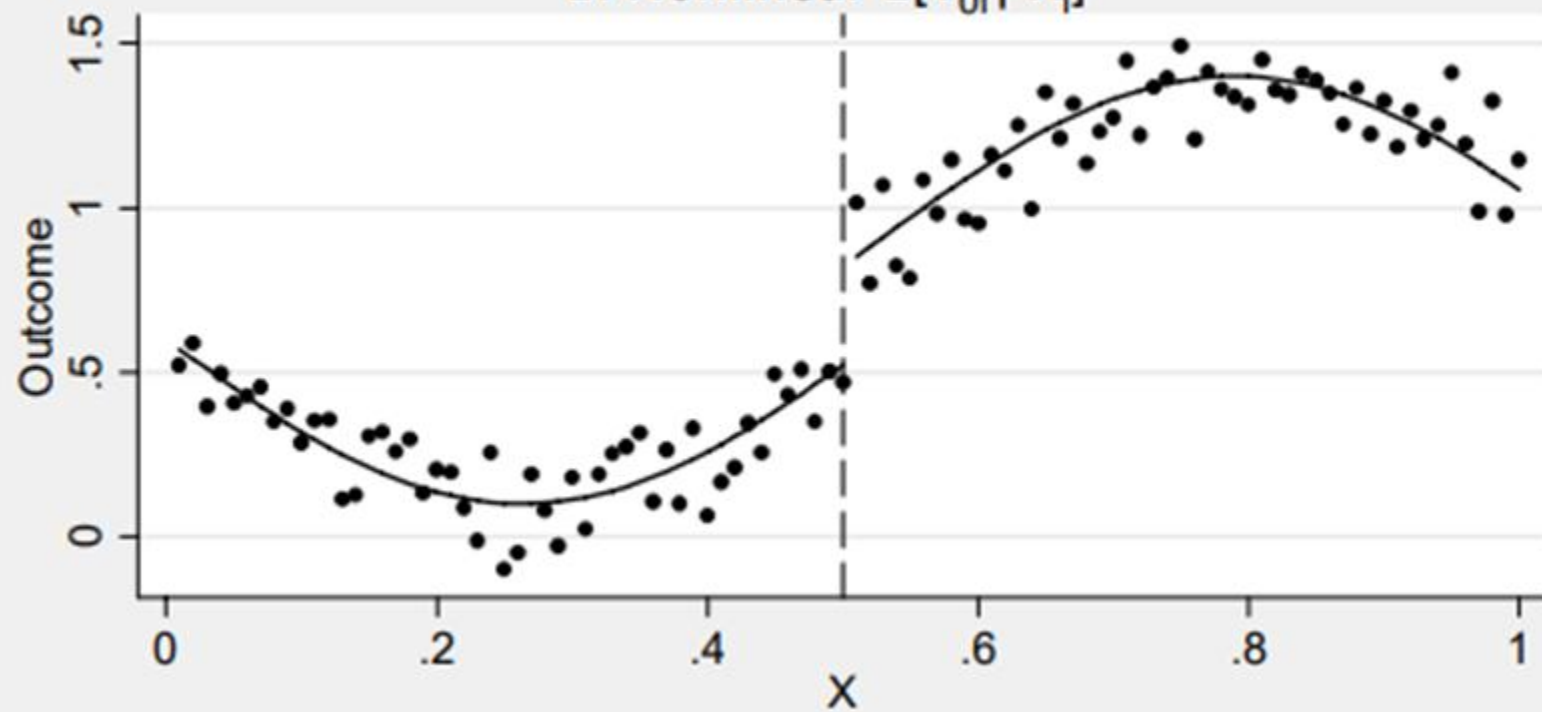


Individual	Outcome	Score	Treatment	Score_c	
1	74	20	1	0	(20-20)
2	46	10	0	-10	(10-20)
3	96	30	1	10	(30-20)
4	54	25	1	5	(25-20)
5	50	22	1	2	(22-20)
6	30	18	0	-2	(18-20)
7	60	12	0	-8	(12-20)
8	40	27	1	7	(27-20)

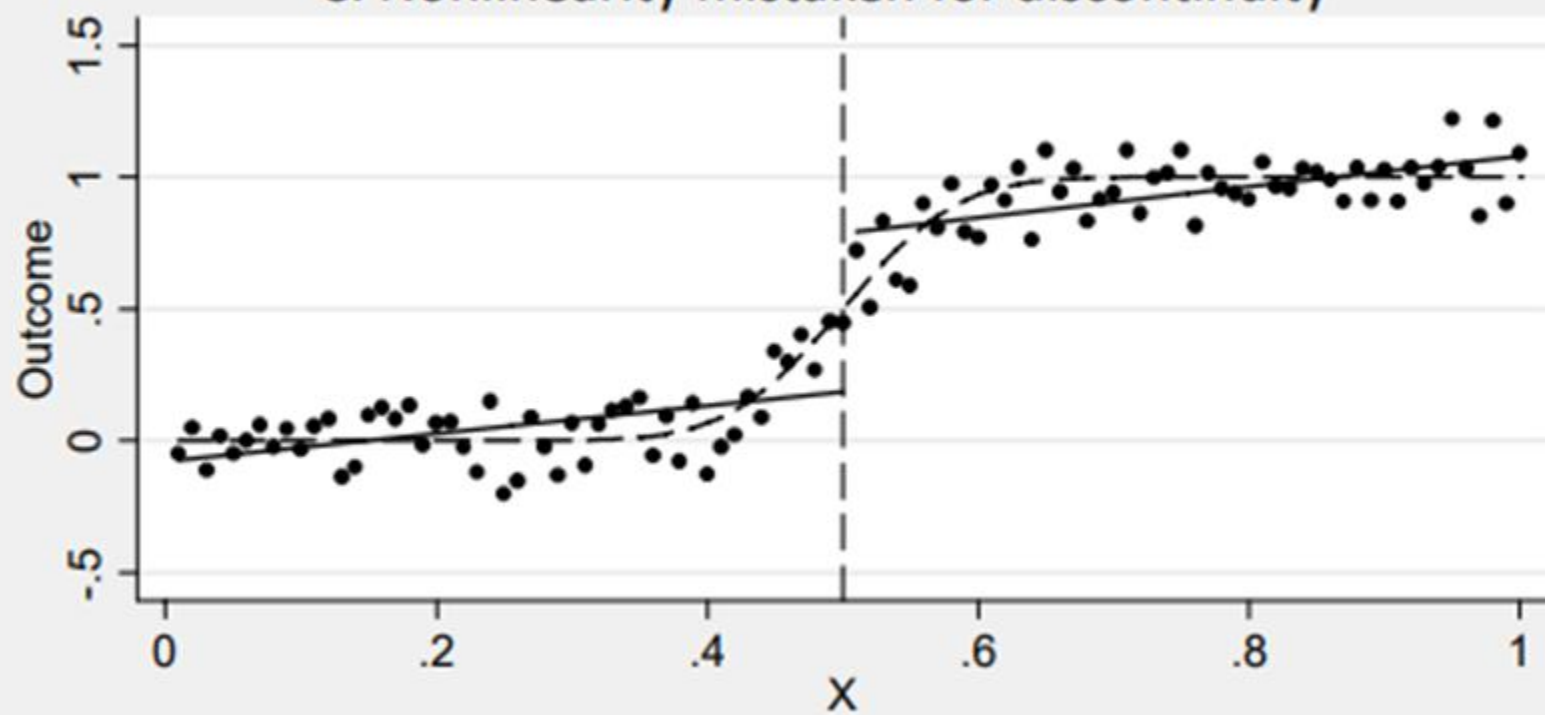
Figure 1.6: Regression discontinuity dataset

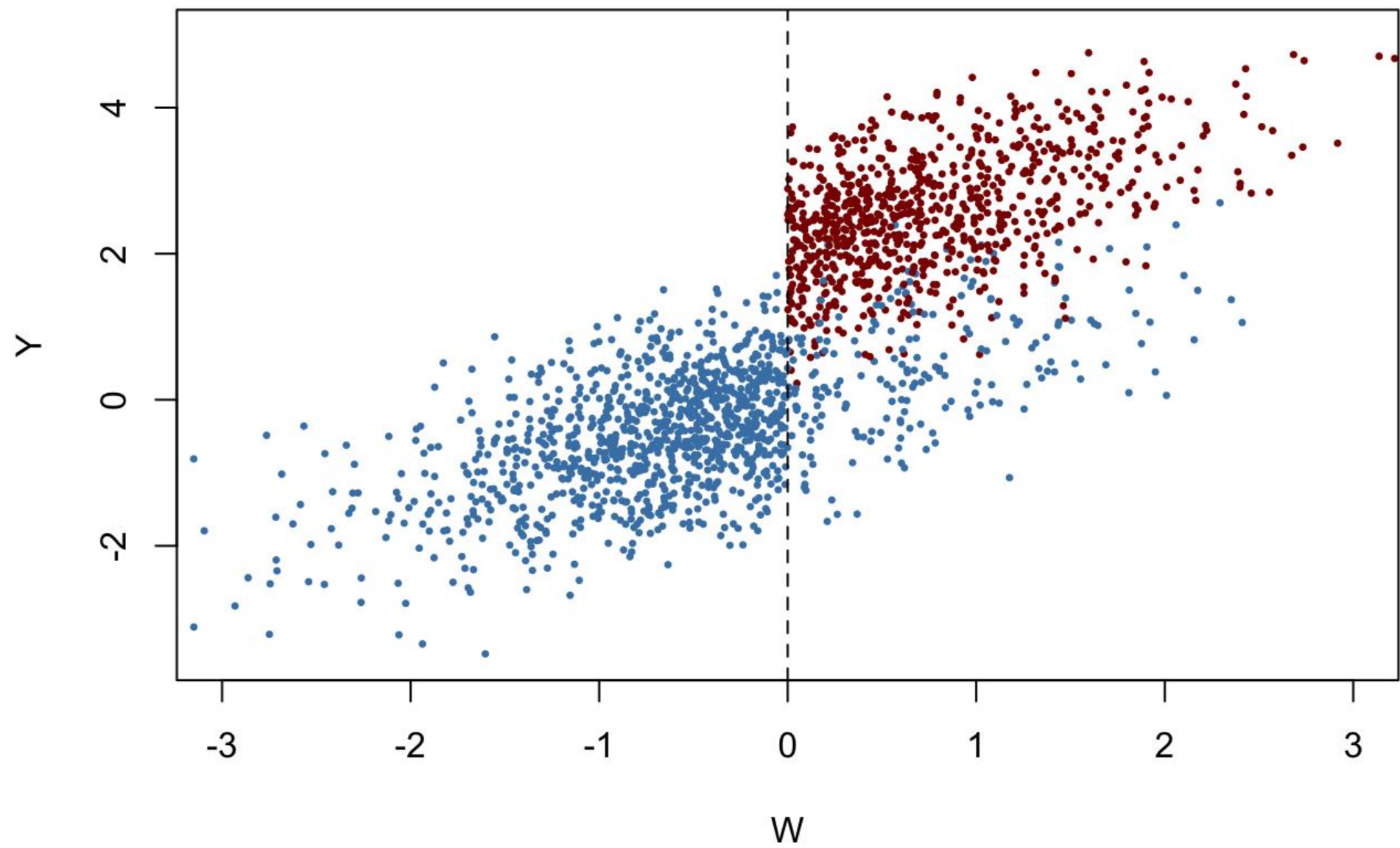


B. Nonlinear $E[Y_{0i} | X_i]$



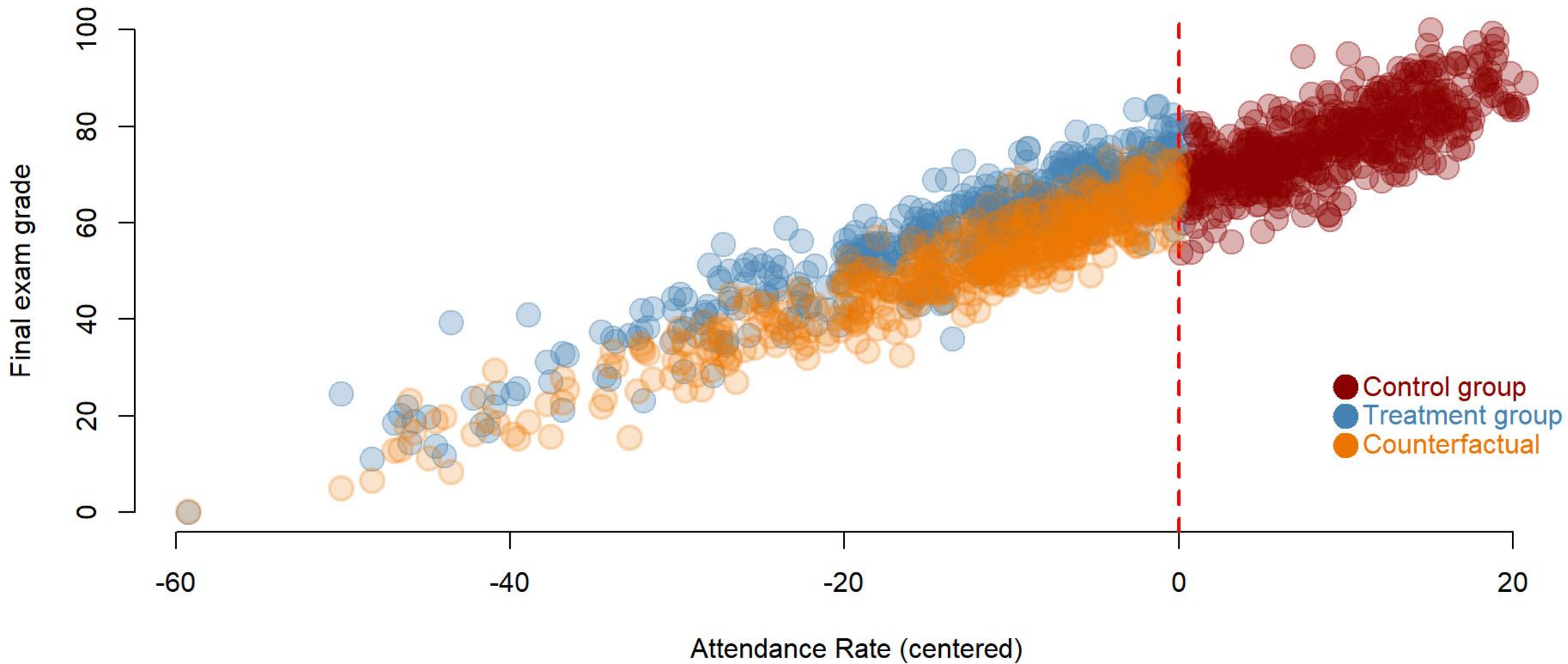
C. Nonlinearity mistaken for discontinuity





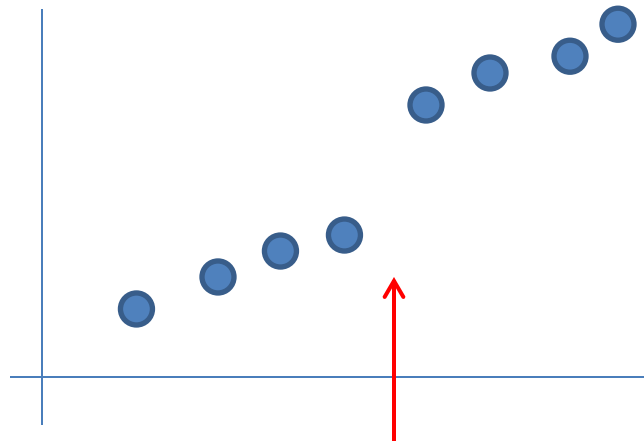
COMPARISON OF METHODS

REGRESSION DISCONTINUITY AND TIME SERIES



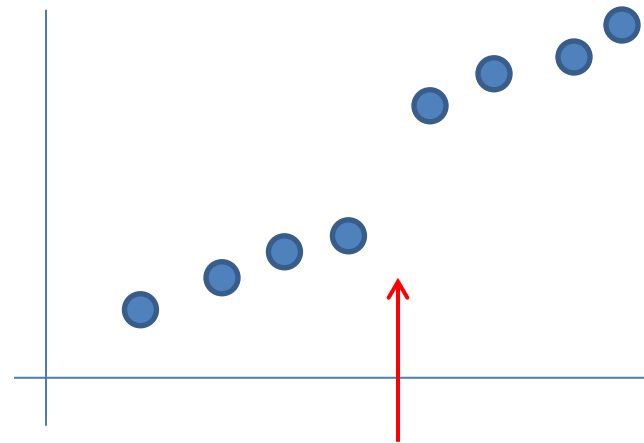
Pay attention to the axis:

Regression Discontinuity



Program

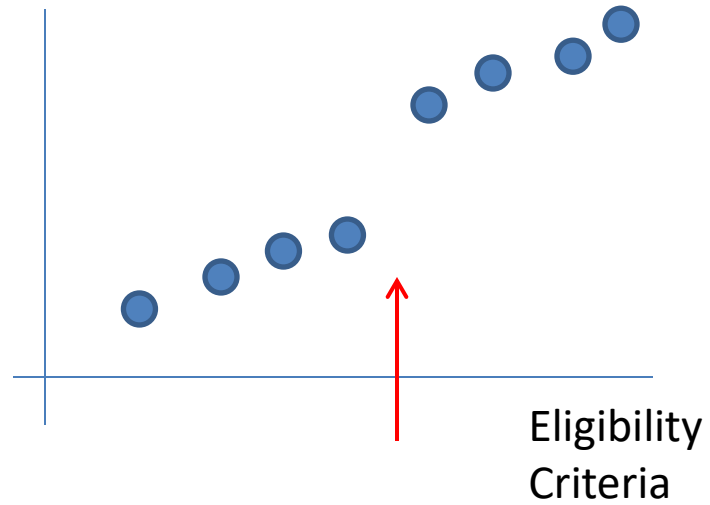
Time Series



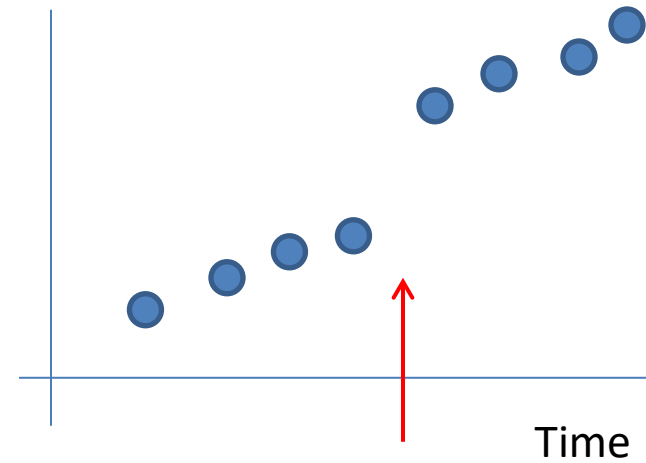
Program

Pay attention to the axis:

Regression Discontinuity



Time Series



Interrupted Time Series Regression Models:

$$Y = b_0 + b_1T + b_2D + b_3P + e$$

Y = outcome

T = time

D = dummy

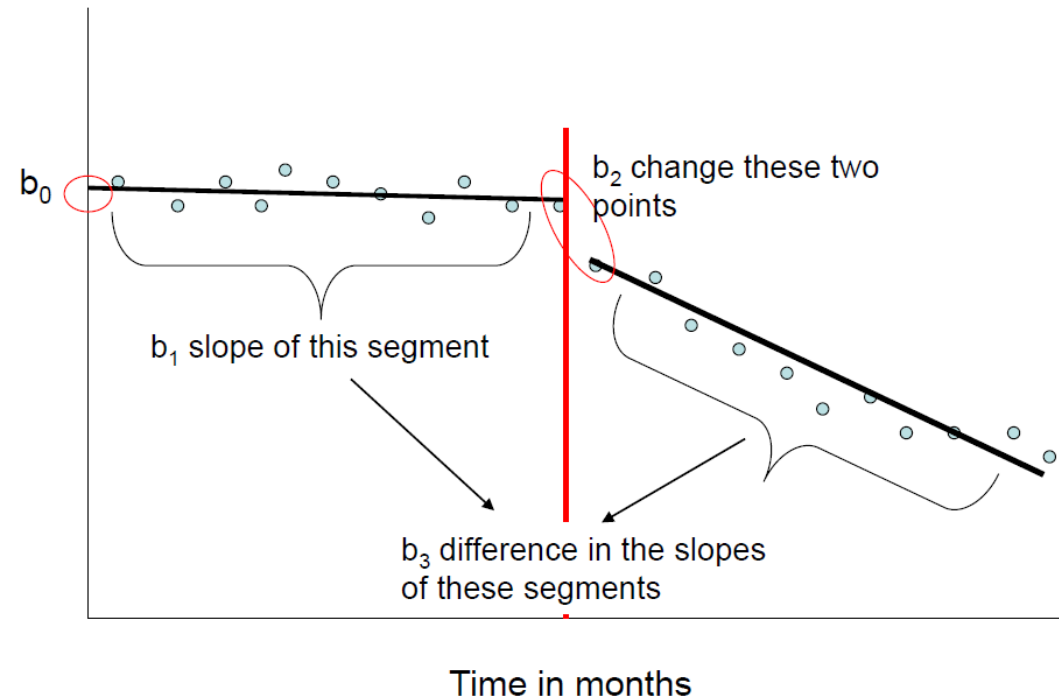
P = time after program starts

Y	T	D	P
17	1	0	0
19	2	0	0
22	3	0	0
24	4	1	1
27	5	1	2
27	6	1	3
30	7	1	4

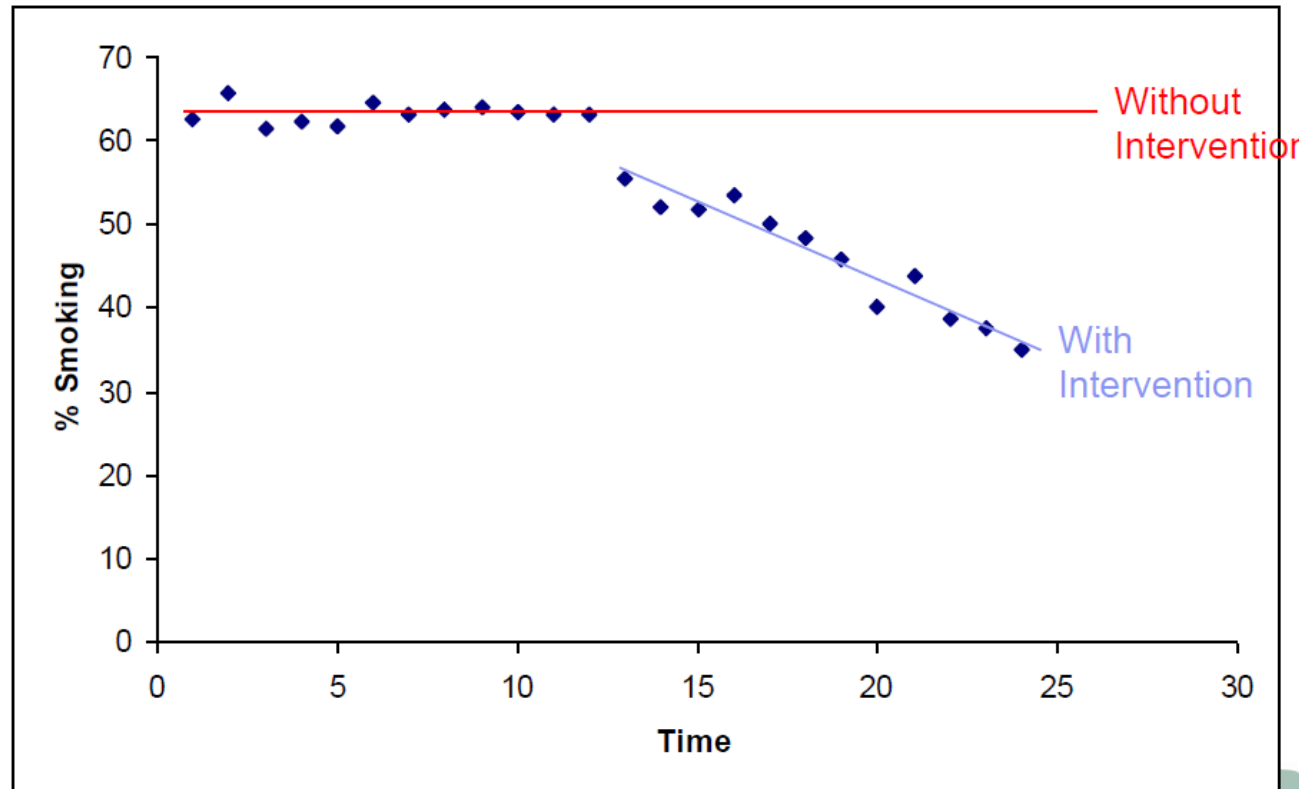
Interrupted Time Series Regression Models:

$$Y = b_0 + b_1T + b_2D + b_3P + e$$

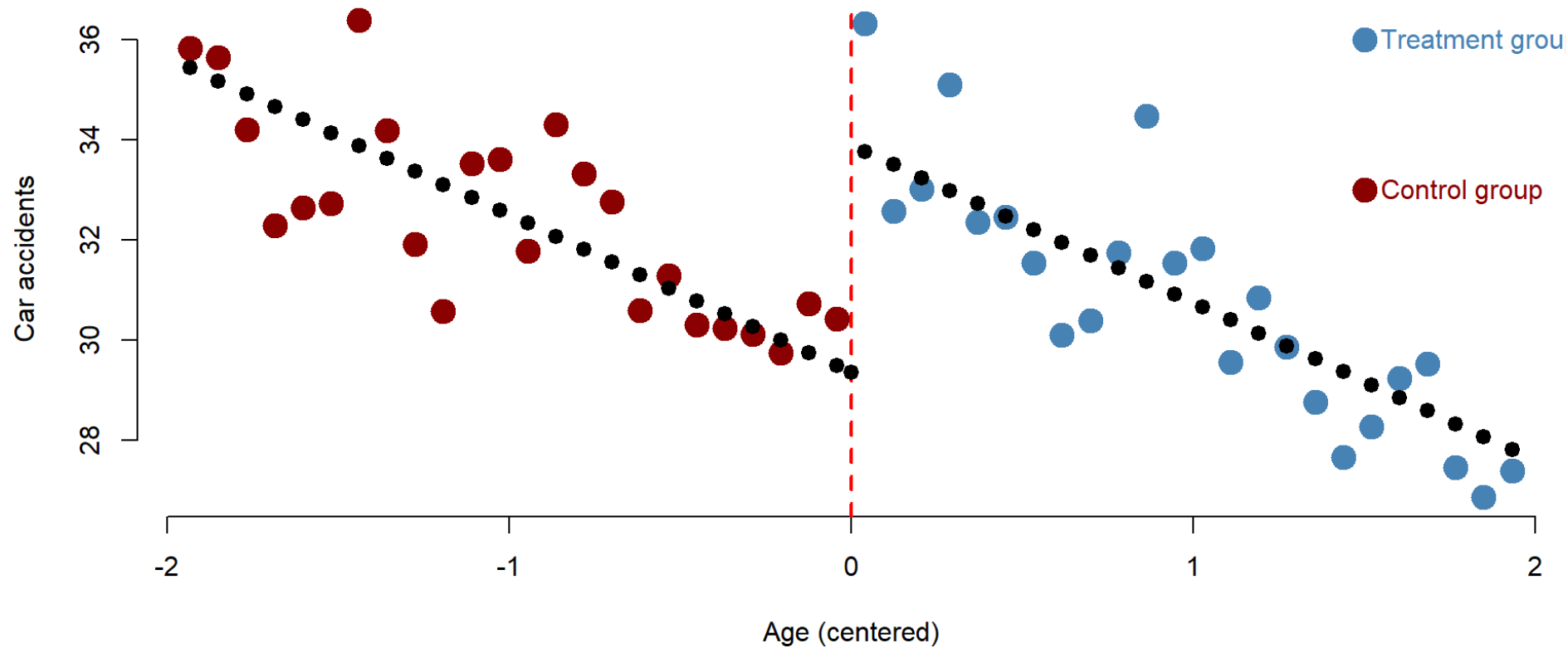
Y	T	D	P
17	1	0	0
19	2	0	0
22	3	0	0
24	4	1	1
27	5	1	2
27	6	1	3
30	7	1	4

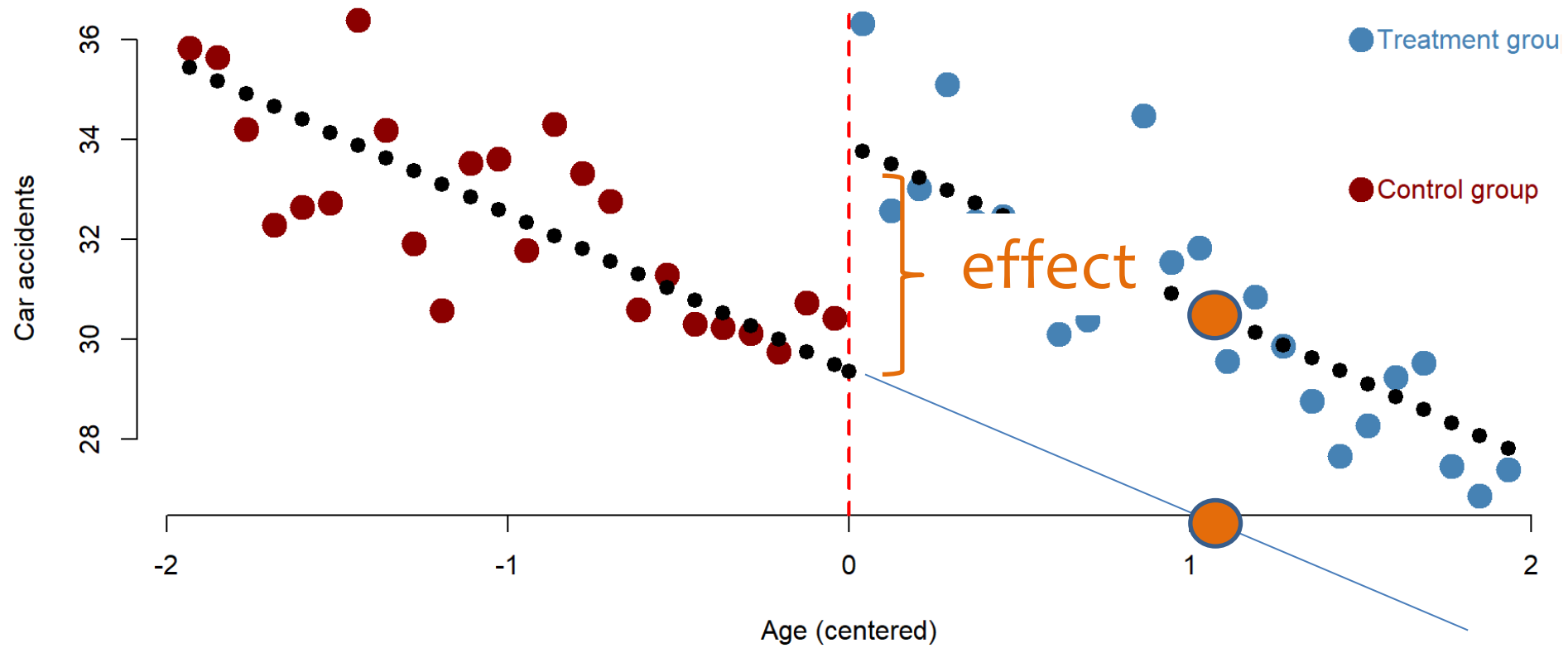


How do you define “effect size”?



How do you define “effect size”?





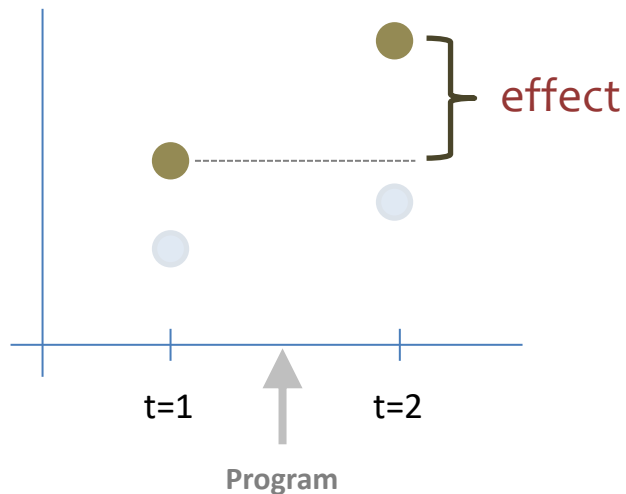
3 VALID COUNTERFACTUALS:

Each variety of counterfactual has a different formula for the program effect estimate.

PRE-POST REFLEXIVE

Estimator

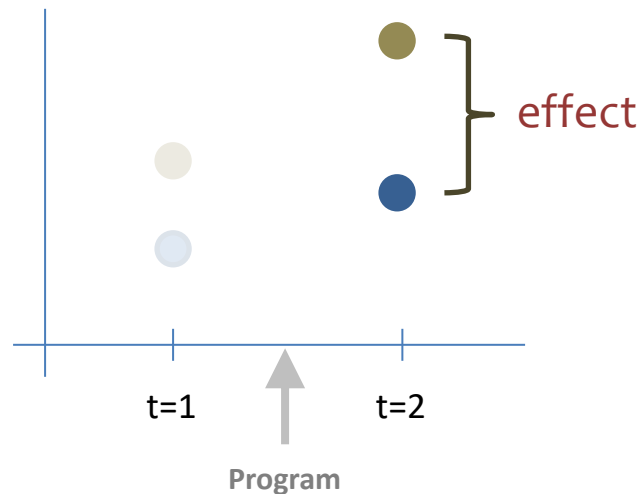
$$\text{effect} = T2 - T1$$



POST-ONLY

Estimator

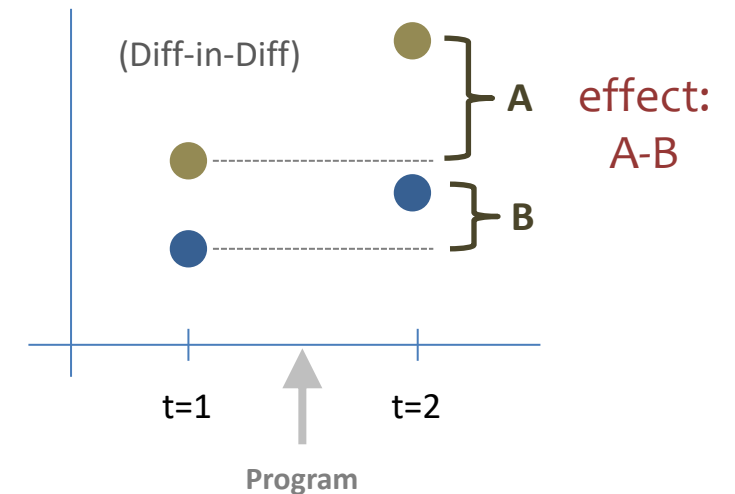
$$\text{effect} = T2 - C2$$



PRE-POST W COMPARISON

Estimator

$$\text{effect} = (T2 - T1) - (C2 - C1)$$



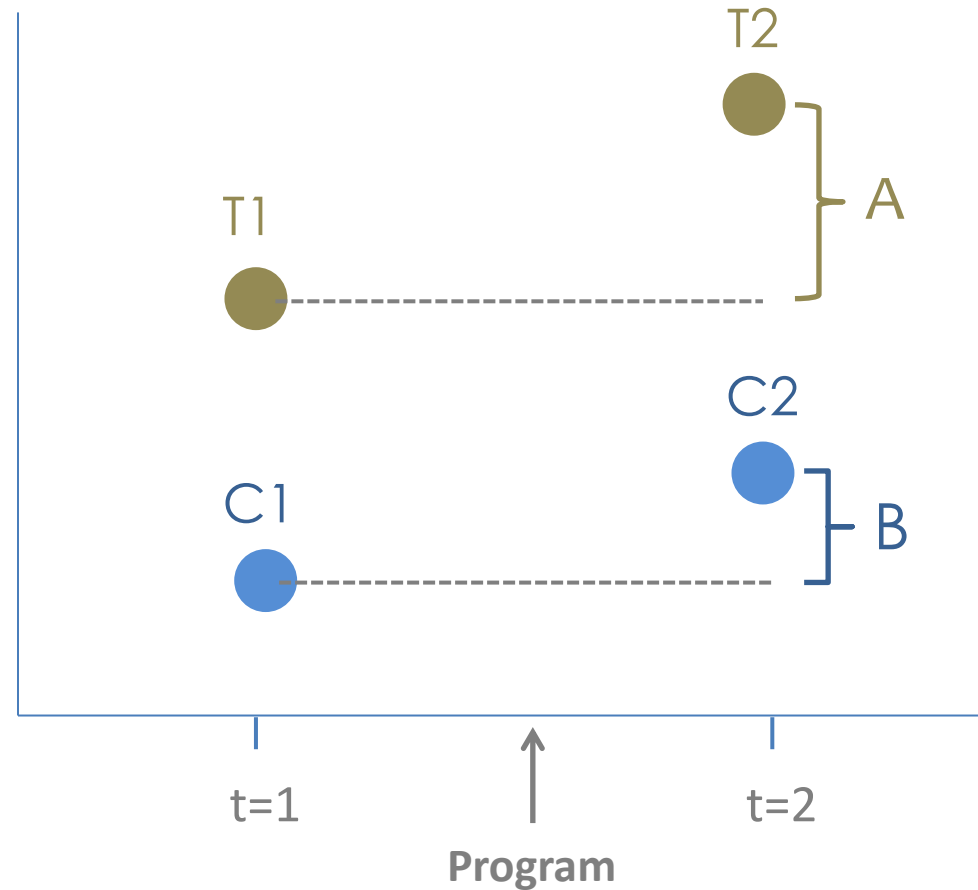
● Treatment Groups, **T1**=before, **T2**=post-program measure

● Control Groups, **C1**=before, **C2**=post-program measure

DIFFERENCE-IN-DIFFERENCE ESTIMATOR

Pre-Post w Comparison
Estimator

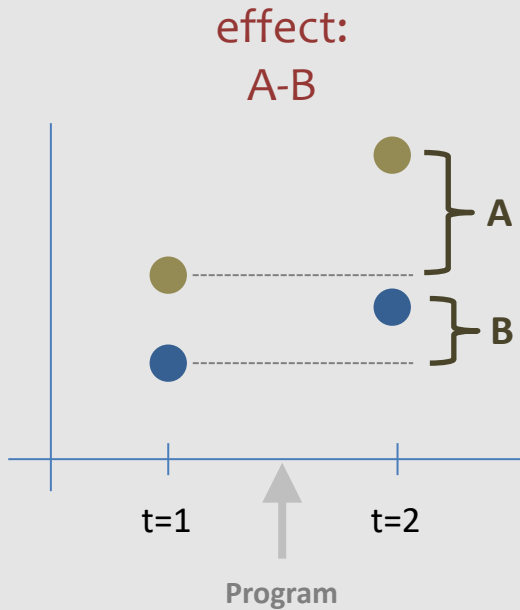
$$\text{effect} = (T2 - T1) - (C2 - C1)$$



effect:
A-B
(Diff-in-Diff)

Accounts for initial difference in groups and secular trends

DIFFERENCE-IN-DIFFERENCE ESTIMATOR



Pre-Post w Comparison

$$\text{effect} = (T2-T1) - (C2-C1)$$

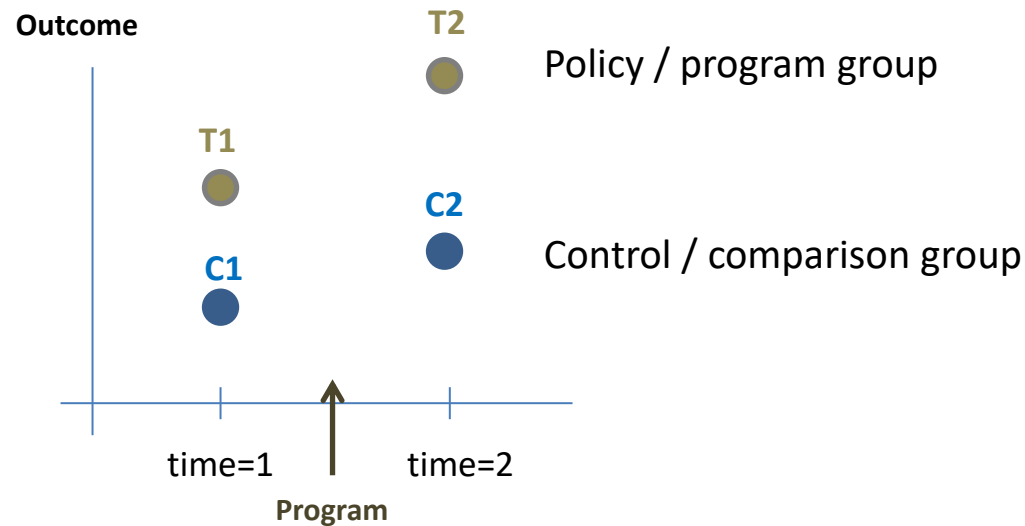


Gains during the
program period by
treatment group



Gains independent
of the program
(secular trends)

"Control" Versus "Comparison" Groups



The term “control group” typically refers to a group created through randomization, so they are assumed to be identical. We sometimes use the terminology “comparison group” to differentiate these cases. The control group represents the counterfactual, the comparison group does not (it just captures trend in this case).

Single Difference:

$$\text{Effect} = T2 - T1$$

$$\text{Effect} = T2 - C2$$

Difference in Difference:

$$\text{Effect} = (T2 - T1) - (C2 - C1)$$

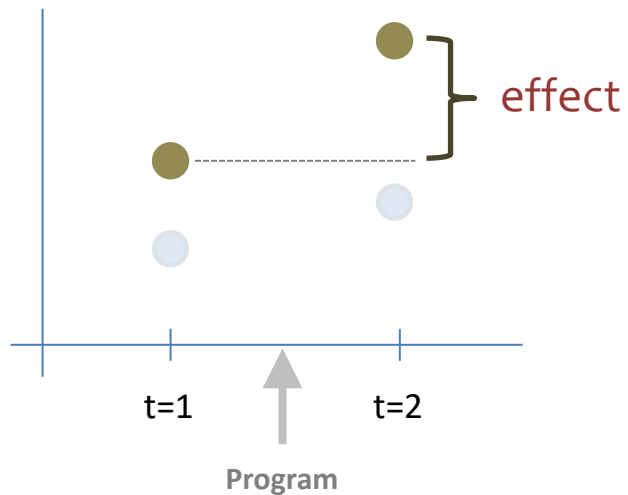
trend

2 REMAINING COUNTERFACTUALS:

PRE-POST REFLEXIVE

Estimator

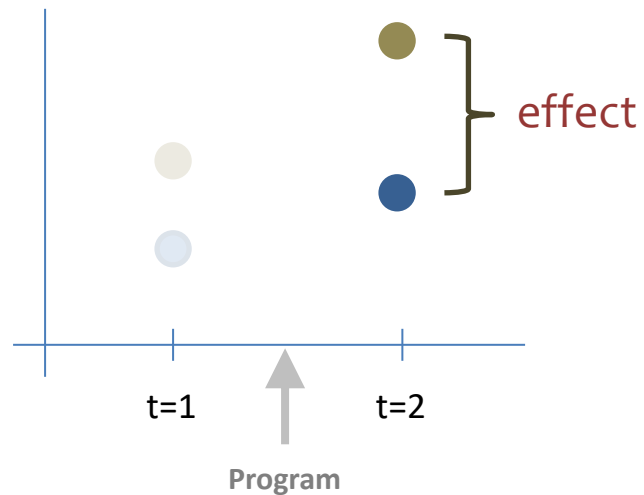
$$\text{effect} = T2 - T1$$



POST-ONLY

Estimator

$$\text{effect} = T2 - C2$$



PRE-POST W COMPARISON

Estimator

$$\text{effect} = (T2 - T1) - (C2 - C1)$$

